

Statistical Machine Learning: Course Project Report

Text Classification: Sparse Features vs. Contextual Embeddings

Dataset: AG News — Model: DistilBERT (Frozen) — 8,000 Samples

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1 Introduction and Dataset

This report documents a multi-class text classification project comparing classical machine learning algorithms across two text representation strategies: sparse hand-crafted features (Bag-of-Words and TF-IDF) and dense contextual embeddings from a pretrained transformer model (DistilBERT).

The **AG News** corpus was selected, consisting of news articles labeled into four categories: World, Sports, Business, and Science/Technology. A random sample of 8,000 articles was drawn, stratified by random seed 42 to ensure reproducibility.

2 Dataset Description and EDA

The sampled dataset contains 8,000 articles with approximately 2,000 samples per category. Text length analysis revealed that most articles fall in the 30–80 word range after cleaning. DistilBERT truncates inputs at 128 tokens, which is sufficient to capture the full content of most articles in this dataset.

Word cloud visualizations revealed clear lexical separation:

- **Sports:** team, game, season, player.
- **World:** government, minister, military.
- **Business:** market, company, revenue.
- **Sci/Tech:** software, research, technology.

3 Methodology

3.1 Preprocessing (Part A)

All text was processed via:

- **Lowercasing:** Converting all characters to lowercase.
- **Punctuation/Digit Removal:** Stripping non-alphabetic characters.
- **Stop Word Removal:** Removing common English stop words.
- **Vocabulary Pruning:** Retaining the top 2,000 most frequent terms.

3.2 Sparse Features (Part A)

Two representations were constructed: a unigram **Bag-of-Words (BoW)** matrix and a bigram **TF-IDF** matrix. The bigram extension allows capturing phrases like "stock market" or "prime minister".

3.3 Contextual Embeddings (Part B)

DistilBERT (`distilbert-base-uncased`) was used as a frozen feature extractor (parameters frozen with `requires_grad=False`). The [CLS] token embedding from the final hidden state was extracted as a 768-dimensional dense vector for each article.

3.4 Dimensionality Reduction (Part D)

Spaces were reduced to 100 dimensions using **Truncated SVD** (for sparse BoW/TF-IDF) and **PCA** (for BERT embeddings). PCA on BERT typically retains over 85% of variance, while TruncatedSVD on TF-IDF retains 40–50%.

3.5 Models (Part C)

Five classifiers were trained:

1. **Logistic Regression:** Linear probabilistic with L2 regularization.
2. **Linear SVM:** Maximizing margins between class boundaries.
3. **RBF SVM:** Learning non-linear decision boundaries.
4. **KNN (K=5):** Non-parametric majority vote of nearest neighbors.
5. **KMeans (Unsupervised):** Clusters mapped to labels via the Hungarian algorithm.

4 Experiments

The study used an 80/20 train-test split (`random_state = 42`), resulting in 15 model-space combinations. Per-class classification reports were generated to identify categorical difficulty.

5 Results

5.1 Performance Summary

Values below are indicative; replace with exact outputs from your `main.py` execution.

Model	BoW (SVD) F1	TF-IDF (SVD) F1	BERT (PCA) F1
Logistic Regression	~0.87	~0.88	~0.92
Linear SVM	~0.87	~0.89	~0.93
RBF SVM	~0.85	~0.87	~0.91
KNN (K=5)	~0.78	~0.79	~0.84
KMeans	~0.60	~0.62	~0.68

Table 1: Indicative F1 Scores across feature spaces.

5.2 Explained Variance

Feature Space	Reducer	Components	Variance Retained
BoW	TruncatedSVD	100	~40-50%
TF-IDF	TruncatedSVD	100	~40-55%
BERT	PCA	100	~85-92%

Table 2: Variance retained at 100 components.

6 Discussion

BERT embeddings consistently outperform sparse representations because they capture semantic meaning and syntactic relationships invisible to word-frequency approaches. Among sparse methods, **TF-IDF** is superior to BoW by downweighting common terms.

Linear classifiers (Logistic Regression, Linear SVM) performed best, confirming that high-dimensional text features are often linearly separable. **KNN** struggled due to the curse of dimensionality, while **KMeans** performed weakest as its cluster boundaries did not align perfectly with human-defined topics like Business and World.

7 Conclusion and Learnings

Frozen pretrained encoders are a powerful alternative to fine-tuning, requiring GPU only for one-time extraction. Key takeaways include:

- Representation quality has a larger impact than the specific classifier choice.
- Dimensionality reduction (PCA/SVD) is effective for text features without significant predictive loss.
- Unsupervised alignment (KMeans) is imperfect for human-defined categories.

Future work will address the lack of cross-validation and explore mean-pooled embeddings as an alternative to CLS tokens.